

Chess position difficulty estimation

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ABSTRACT

In this paper we implement and compare various procedures for assigning an ELO estimate to any chess position and for ranking the positions without providing numerical outputs.

KEYWORDS

chess, classification, regression, ranking

1 INTRODUCTION

In this paper we compare various methods for determining the difficulty of a given chess position for human players, using chess engine evaluations and other attributes of the position. This problem can be defined in the following three ways:

- classification: the model classifies positions into pre-defined classes (e.g. easy/medium/hard), or binned ELO ratings.
- regression: the model directly predicts an estimated ELO rating of the position.
- ranking: the model orders a set of positions from easiest to hardest, or ranks a single position in relation to all positions in the training set.

All the developed methods are available at ¹.

2 RELATED WORK

Recent approaches for estimating the difficulty of a chess position, e.g. [3] are largely based on the deep learning paradigm and the input of these models consists of some encoding of the chess position, but no information that would provide interpretability for human players. In this paper we therefore focus on finding features that achieve good classification score, but also provide a meaningful interpretation regarding the reasons why a position is considered difficult or easy for human players.

Our approaches are based on insights provided by [2] [1] [4]. These studies developed numerous attributes that can be obtained by performing a chess engine search of certain depth, starting in the initial position we wish to evaluate. The search is constrained in such a way that only moves which human players would typically assess as meaningful are explored, which aims to provide an estimate of how complex the search tree around the position is.

3 METHODOLOGY

To train and evaluate our models, we use a subset of the Lichess puzzle dataset ² containing 1000 puzzles with ratings ranging between

500 and 3100. For classification we create classes "easy", "medium" and "hard" by splitting this interval equidistantly. The dataset is split into training set and test set with the ratio 80:20. Both sets are standardized to have zero mean and unit variance.

3.1 Evaluation Metrics

For classification problem we use the standard metrics: classification accuracy, F1 score and AUC. For regression we use mean absolute error (MAE) because unlike mean squared error, it has a more intuitive interpretation on this dataset. For ranking we use Kendall τ to determine the accuracy of the model's ranking against the ground truth, which is simply a ranking assigned by ordering the ratings in the dataset.

3.2 Attributes

The original dataset contains no attributes that would allow modeling of perceived difficulty for human players, so we expand it with the following attributes:

- Stockfish parameters for each depth between 1-3 from starting position. These parameters include centipawn evaluations, number of searched nodes, WDL (win/draw/loss) and selective depth.
- material count and piece counts for each side.
- attributes from [4], computed for depth=3, excluding "Seemingly Good" due to ambiguous definition.
- binary attributes for solving success by Stockfish engine with different playing strengths.

3.3 Hierarchical Model

We propose a hierarchical classification/regression model, consisting of a main model, which classifies the position into one of the predetermined equidistant rating bins and a dedicated, more precise, separately trained model for each of the bins.

4 RESULTS

5 CONCLUSIONS

Future work would consist of checking if these approaches are suitable also for strategic/nondescript positions, rather than just puzzles. Here, the main challenge is in assigning the ground truth

¹<https://github.com/bajicluka01/chess-puzzle-evaluation/tree/main>

²<https://database.lichess.org/#puzzles>

Table 1: Classification accuracies.

model	CA	F1	AUC
XGBoost	0.485	0.485	0.485
Logistic regression	0.705	0.485	0.485
Naive Bayes	0.705	0.485	0.485
kNN	0.705	0.485	0.485
Random Forest	0.705	0.485	0.485
Gradient Boosting	0.705	0.485	0.485
Neural Network	0.705	0.485	0.485
SGD	0.705	0.485	0.485
SVM	0.705	0.485	0.485

Table 2: Regression accuracies.

model	MAE
XGBoost	0.485
Linear regression	0.705
Neural Network	0.705
Bayes Ridge	0.705
SGD	0.705

Table 3: Ranking accuracies.

model	NDCG	τ
XGBRanker	0.485	0
Logistic regression	0.705	0

ELO to these positions, since consensus among human players is more difficult to achieve.

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